DSA0605 – Data Handling and Visualization

CO1-AT3-Visualization Design Exercise

Name: P. Siddardha

Reg.no: 192424271

Question 1: Integrated Visualization of Retail Branch Performance

Objective
Design a visualization that simultaneously represents Sales, Profit, Number of Customers, Product Category, and Region.

1.1 R program
The most suitable design is a bubble scatter plot with Sales on the x-axis, Profit on the y-axis, Customer count represented by bubble size, Product Category by colour, and Region by shape. Since sales and profit can vary substantially, a symmetric log transformation is used for Profit to retain negative profit values while making the distribution easier to inspect.

```
# Question 1: Retail Branch Performance

# Install once if required:
# install.packages("ggplot2")
# install.packages("scales")

library(ggplot2)
library(scales)

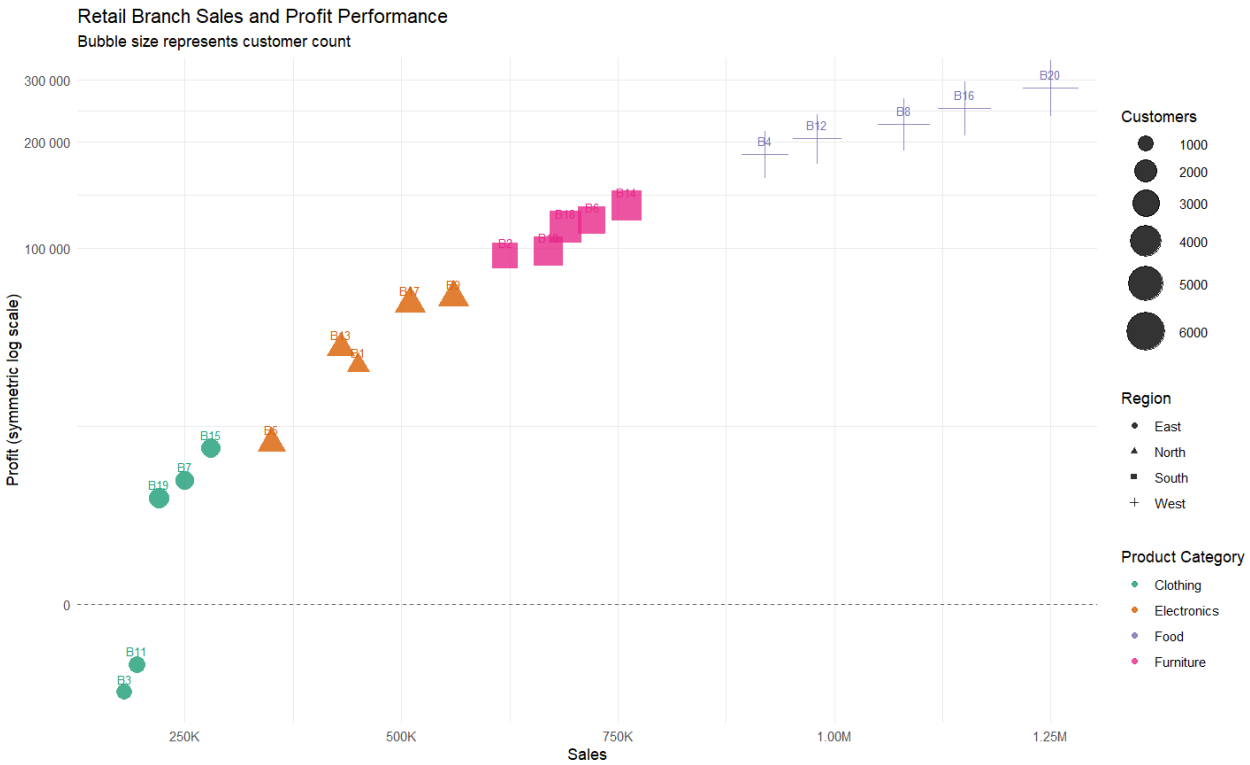
retail <- data.frame(
  Branch = paste0("B", 1:20),
  Region = c(
    "North", "South", "East", "West",
    "North", "South", "East", "West",
    "North", "South", "East", "West",
    "North", "South", "East", "West",
    "North", "South", "East", "West"
  ),
  Category = c(
    "Electronics", "Furniture", "Clothing", "Food",
    "Electronics", "Furniture", "Clothing", "Food",
    "Electronics", "Furniture", "Clothing", "Food",
    "Electronics", "Furniture", "Clothing", "Food",
    "Electronics", "Furniture", "Clothing", "Food"
  ),
  Sales = c(
    450000, 620000, 180000, 920000, 350000,
    720000, 250000, 1080000, 560000, 670000,
```

```
    195000, 980000, 430000, 760000, 280000,
    1150000, 510000, 690000, 220000, 1250000
  ),
  Profit = c(
    45000, 95000, -12000, 185000, 25000,
    120000, 18000, 225000, 72000, 98000,
    -8000, 205000, 51000, 132000, 24000,
    250000, 69000, 115000, 15000, 285000
  ),
  Customers = c(
    1200, 2500, 980, 4200, 1800,
    3000, 1250, 5200, 2100, 3300,
    1100, 4700, 1750, 3600, 1450,
    5600, 2200, 3900, 1600, 6100
  )
)

retail$Region <- factor(retail$Region)
retail$Category <- factor(retail$Category)

ggplot(retail, aes(
  x = Sales,
  y = Profit,
  size = Customers,
  colour = Category,
  shape = Region
)) +
  geom_hline(yintercept = 0, linetype = "dashed",
             colour = "grey40") +
  geom_point(alpha = 0.8) +
  geom_text(aes(label = Branch),
            vjust = -0.9, size = 2.8,
            show.legend = FALSE) +
  scale_x_continuous(
    labels = label_number(scale_cut = cut_short_scale())
  ) +
  scale_y_continuous(
    trans = scales::pseudo_log_trans(sigma = 10000),
    labels = label_number()
  ) +
```

```
  scale_size_area(
    max_size = 12,
    name = "Customers"
  ) +
  scale_colour_brewer(
    palette = "Dark2",
    name = "Product Category"
  ) +
  labs(
    title = "Retail Branch Sales and Profit Performance",
    subtitle = "Bubble size represents customer count",
    x = "Sales",
    y = "Profit (symmetric log scale)",
    shape = "Region"
  ) +
  theme_minimal() +
  theme(legend.position = "right")
```



1.2 Explain the selected aesthetic mappings.

Variable	Aesthetic	Justification
Sales	X-axis position	Shows branch sales magnitude
Profit	Y-axis position	Shows profitability, including losses

Customers	Bubble size	Larger circles represent more customers
Product Category	Colour	Distinguishes Electronics, Furniture, Clothing, and Food
Region	Shape	Distinguishes North, South, East, and West
Branch	Text label	Identifies individual branches

Position is used for Sales and Profit because it supports more accurate comparisons than colour or bubble size alone.

1.3 Justify the scale transformation.

The profit data contains both positive and negative values, ranging from a loss of ₹8,000 to a profit of ₹285,000.

A conventional logarithmic transformation cannot directly represent negative and zero values. The pseudo_log_trans() transformation handles negative values, zero, and positive values while behaving approximately logarithmically for larger magnitudes.

Sales remains on a linear scale because its values are all positive and its range is manageable.

1.4 Interpret the visualization.

- Branch B20 has the highest sales of ₹1,250,000 and the highest profit of ₹285,000.
- Branch B3 records a loss of ₹12,000, despite making ₹180,000 in sales.
- Branch B12 generates ₹980,000 in sales and ₹205,000 in profit.
- Branch B19 has relatively low sales of ₹220,000 and a positive profit of ₹15,000.

The bubble sizes indicate differences in customer counts, while colour and shape allow comparisons across categories and regions.

1.5 Justify the overall visualization design.

The bubble scatter plot combines all five variables in a single chart without requiring separate axes for every measure. Position provides the most precise encoding for Sales and Profit, bubble area represents customer count, colour distinguishes categories, and shape distinguishes regions. The symmetric log transformation preserves loss-making branches and reduces the visual dominance of large profits. Branch labels support identification, while legends explain each aesthetic mapping.

Conclusion: This integrated visualization supports comparisons of branch performance, customer reach, product category, and geographical region while retaining important information about losses.

Question 2: Cartesian and Polar Coordinate Visualization of Energy Generation

Objective

Visualize monthly electricity generation from Coal, Gas, Solar, Wind, and Hydro using Cartesian and polar coordinate systems, compare their effectiveness, and justify the design decisions.

2.1 R program using Cartesian coordinates

A multi-series line chart is suitable because the data represents monthly changes across five energy sources.

```
# Question 2.1: Cartesian Coordinate System

library(ggplot2)

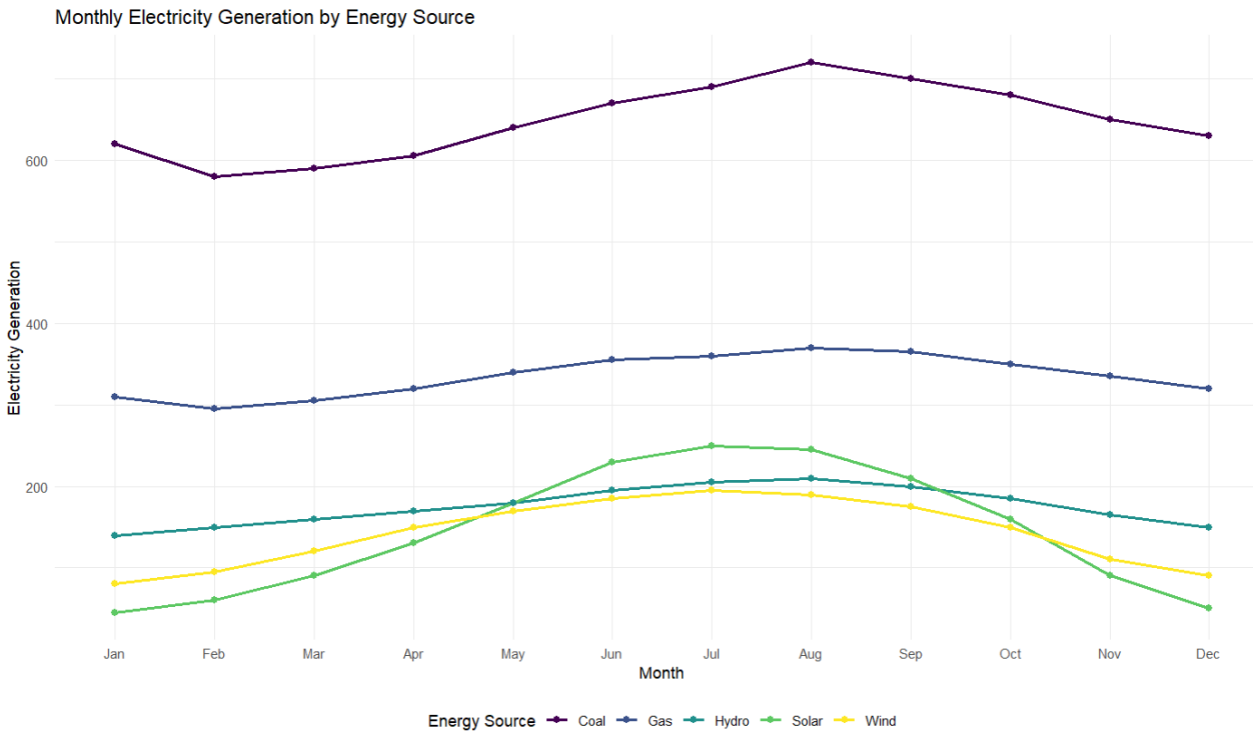
energy <- data.frame(
  Month = month.abb,
  Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),
  Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),
  Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),
  Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),
  Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)
)

# Convert to long format
energy_long <- data.frame(
  Month = rep(energy$Month, 5),
  Source = rep(c("Coal", "Gas", "Solar", "Wind", "Hydro"),
               each = 12),
  Generation = c(
    energy$Coal, energy$Gas, energy$Solar,
    energy$Wind, energy$Hydro
  )
)

energy_long$Month <- factor(
  energy_long$Month,
  levels = month.abb,
  ordered = TRUE
)

ggplot(energy_long, aes(
  x = Month,
  y = Generation,
  colour = Source,
  group = Source
)) +
```

```
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_colour_viridis_d(option = "D") +
  labs(
    title = "Monthly Electricity Generation by Energy Source",
    x = "Month",
    y = "Electricity Generation",
    colour = "Energy Source"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



Explanation: geom_line() shows monthly trends, geom_point() identifies each monthly observation, and colour distinguishes the five energy sources. The Cartesian coordinate system makes the magnitudes and changes in generation easy to compare.

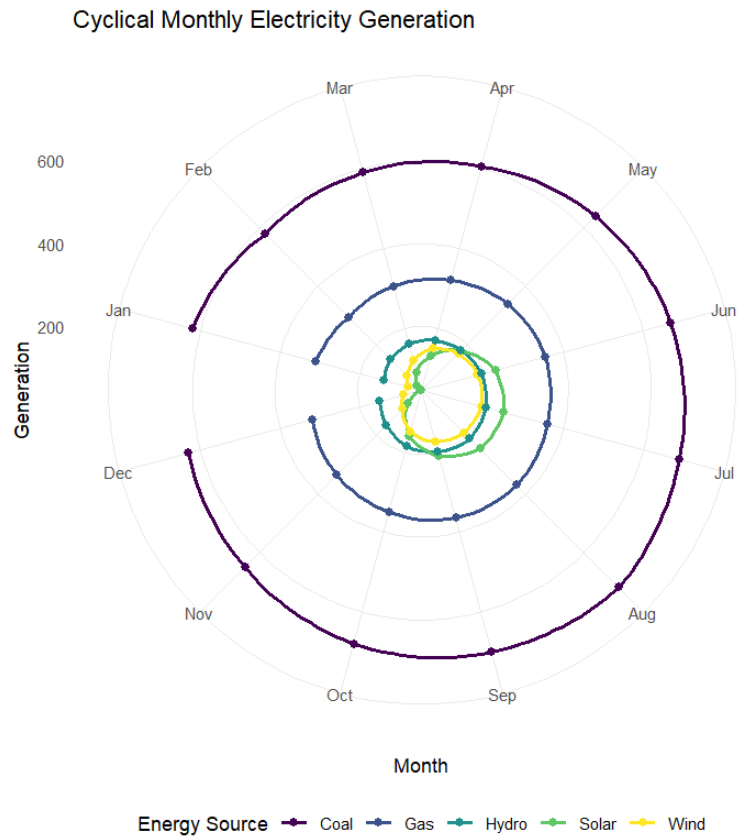
2.2 R program using polar coordinates

```
# Question 2.2: Polar Coordinate System

library(ggplot2)

ggplot(energy_long, aes(
```

```
  x = Month,
  y = Generation,
  colour = Source,
  group = Source
)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  coord_polar(theta = "x", start = -pi / 2) +
  scale_colour_viridis_d(option = "D") +
  labs(
    title = "Cyclical Monthly Electricity Generation",
    x = "Month",
    y = "Generation",
    colour = "Energy Source"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



Explanation: coord_polar(theta = "x") arranges months around a circle, while the radial position represents generation. The colours identify the five energy sources.

This is a circular representation of the annual cycle. It is useful for visualizing seasonality, although connecting December to January visually should not be interpreted as proof that the two values are identical or that a trend is continuous across the boundary.

2.3 Compare the effectiveness of both visualizations.

Feature	Cartesian chart	Polar chart
Layout	Rectangular axes	Circular layout
Monthly comparisons	Easy and precise	Less precise
Trend identification	Clear	Can be harder with overlapping lines
Seasonal pattern	Visible	Emphasizes annual cyclicity
Reading exact values	Easier	More difficult
Best application	Analytical comparison	Seasonal presentation

2.4 Determine which coordinate system provides better analytical insight.

The Cartesian coordinate system provides better analytical insight for this dataset because it makes it easier to compare the generation values of five energy sources across all twelve months. The polar system is useful when the principal goal is to emphasize recurring annual patterns, but radial positions and overlapping lines make exact comparisons harder.

2.5 Explain and justify the design decisions.

- Line chart: Appropriate for showing changes across the twelve months.
- Month on the x-axis: Preserves chronological order.
- Generation on the y-axis: Supports numerical comparisons.
- Colour mapped to energy source: Separates the five series.
- Viridis palette: Provides perceptually ordered, distinguishable colours.
- Cartesian coordinates: Make increases, decreases, and differences between sources easier to evaluate.

The data shows that Coal generation is consistently the highest among the five sources. Solar generation rises from 45 in January to a peak of 250 in June, then declines to 50 in December. Wind generation also rises toward the middle of the year.

Conclusion: Cartesian coordinates are better for detailed analytical comparisons, whereas polar coordinates are useful for communicating annual seasonality.

Question 3: Climate Visualization Using Appropriate Colour Scales

Objective

Represent Temperature, AQI, Rainfall, and Humidity for twenty Indian cities using appropriate colour scales and justify the choice of palette.

3.1 R program to visualize all four climate variables

A standardized heatmap is suitable because the four variables use different units and have very different numerical ranges. Standardization allows each city to be compared with other cities for the same variable.

```
# Question 3.1: Climate Data Heatmap

library(ggplot2)
library(scales)

climate <- data.frame(
  City = c(
    "Delhi", "Mumbai", "Chennai", "Kolkata", "Hyderabad",
    "Bangalore", "Pune", "Ahmedabad", "Jaipur", "Lucknow",
    "Bhopal", "Patna", "Goa", "Kochi", "Nagpur", "Indore",
    "Surat", "Visakhapatnam", "Coimbatore", "Mysuru"
  ),
  Temperature = c(
    44,35,38,40,39,31,33,43,45,39,
    38,40,32,30,41,37,36,34,29,28
  ),
  Humidity = c(
    22,84,78,72,48,66,58,26,18,42,
    36,45,89,91,30,34,76,82,68,64
  ),
  Rainfall = c(
    6,280,220,190,85,120,95,2,1,35,
    48,72,360,410,12,20,180,240,130,145
  ),
  AQI = c(
    365,110,125,180,160,72,84,290,320,240,
    175,220,58,52,270,190,145,98,60,55
  )
)

# Standardize each variable separately
climate_z <- climate

climate_z[2:5] <- lapply(
  climate[2:5],
  function(x) as.numeric(scale(x))
```

```
)

# Convert to long format
climate_long <- data.frame(
  City = rep(climate$City, 4),
  Variable = rep(
    c("Temperature", "Humidity", "Rainfall", "AQI"),
    each = nrow(climate)
  ),
  StandardScore = c(
    climate_z$Temperature,
    climate_z$Humidity,
    climate_z$Rainfall,
    climate_z$AQI
  )
)

climate_long$City <- factor(
  climate_long$City,
  levels = rev(climate$City)
)

climate_long$Variable <- factor(
  climate_long$Variable,
  levels = c("Temperature", "AQI", "Rainfall", "Humidity")
)

ggplot(climate_long, aes(
  x = Variable,
  y = City,
  fill = StandardScore
)) +
  geom_tile(color = "white", linewidth = 0.3) +
  scale_fill_viridis_c(
    option = "C",
    name = "Standardized\nScore"
  ) +
  labs(
    title = "Climate Conditions Across Indian Cities",
    subtitle = "Standardized values (z-scores)",
```

```
  x = "Climate Variable",
  y = "City"
) +
theme_minimal() +
theme(
  axis.text.y = element_text(size = 8),
  axis.text.x = element_text(angle = 25, hjust = 1)
)
```



3.2 Explain the aesthetic mappings.

Variable	Visual mapping	Purpose
City	Y-axis	Identifies each city
Climate variable	X-axis	Separates Temperature, AQI, Rainfall, and Humidity
Standardized value	Fill colour	Communicates relative magnitude

The heatmap provides a compact view of all eighty city-variable combinations.

3.3 Explain the scale transformation.

Temperature, AQI, Rainfall, and Humidity have different units and ranges. For example, rainfall ranges from 1 to 410, while temperature ranges from 28 to 45.

The program standardizes each variable using a z-score:

$$ z=\frac{x-\bar{x}}{s} $$

where x is an observation, $\bar{x}$ is the variable's mean, and s is its standard deviation.

- A positive z-score means the value is above that variable's mean.
- A negative z-score means the value is below that variable's mean.

- A z-score near zero means the value is near the mean.

This is a standardization transformation, not a change to the original measurements.

3.4 Suggest a suitable colour palette and justify its effectiveness.

The Viridis palette is suitable because it provides perceptually ordered colours and is designed to be accessible to many users with colour-vision deficiencies.

It supports consistent interpretation of low and high standardized values without requiring red-green discrimination. A continuous palette is appropriate because the heatmap represents numerical data rather than unrelated categories.

3.5 Interpret the visualization and explain its limitations.

- Delhi has a high AQI of 365, indicating substantially greater air pollution than cities such as Kochi and Mysuru.
- Jaipur has the highest temperature at 45, while Mysuru has the lowest at 28.
- Kochi has the highest rainfall at 410 and humidity at 91.
- Goa also has high rainfall of 360 and humidity of 89.

The heatmap makes relative differences visible across cities and variables. However, standardized colours do not directly communicate the original units or health thresholds. A high standardized value means high relative to this dataset, not necessarily the worst possible environmental condition. AQI interpretation should use the applicable official AQI scale.

Conclusion: A standardized heatmap with Viridis colour mapping enables compact, accessible comparison of four climate variables with different units. Original numerical values and domain-specific thresholds should accompany it when precise interpretation is required.

Question 4: Nonlinear Axis Transformation for Software Project Statistics

Objective

Design a visualization for twenty software development teams that uses a nonlinear axis transformation to represent highly skewed data and explain its advantages over a linear scale.

4.1 R program using a logarithmic transformation

Lines of Code ranges from 12,000 to 1,800,000, so a logarithmic x-axis is suitable for displaying the large range of project sizes.

```
# Question 4.1: Nonlinear Axis Transformation

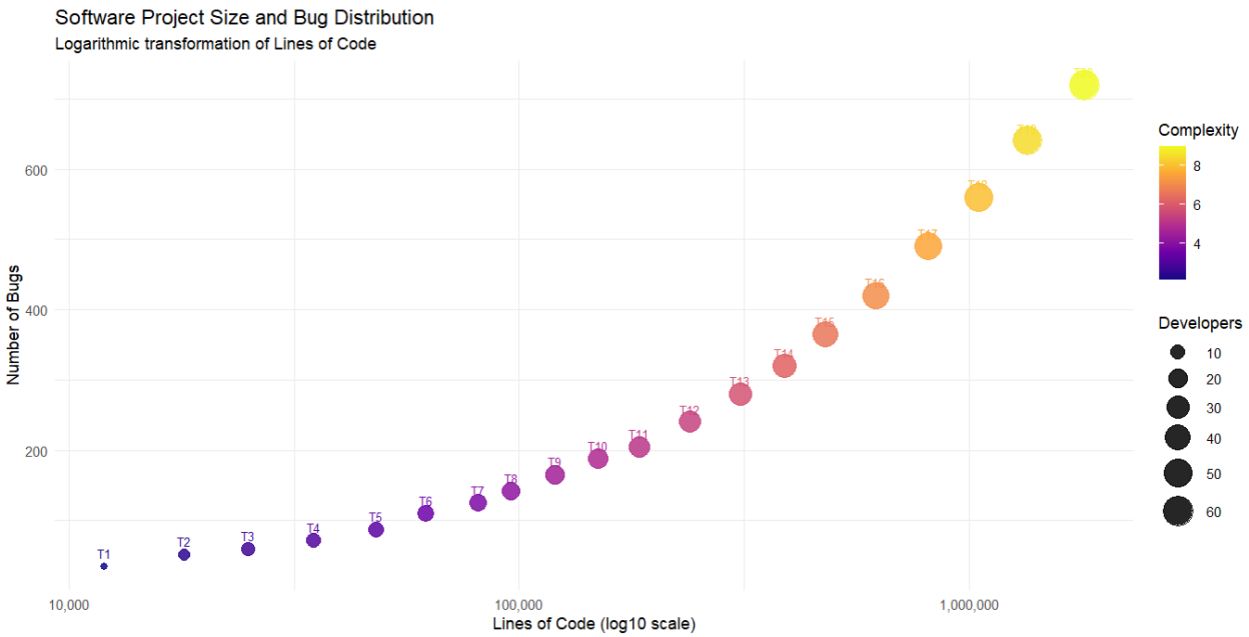
library(ggplot2)
library(scales)

software <- data.frame(
  Team = paste0("T", 1:20),
```

```
  LinesOfCode = c(
    12000,18000,25000,35000,48000,
    62000,81000,96000,120000,150000,
    185000,240000,310000,390000,480000,
    620000,810000,1050000,1350000,1800000
  ),
  Bugs = c(
    35,52,60,72,88,110,125,142,165,188,
    205,240,280,320,365,420,490,560,640,720
  ),
  Developers = c(
    5,7,9,10,12,14,16,18,20,22,
    25,28,30,34,38,42,46,50,55,60
  ),
  Complexity = c(
    2.1,2.4,2.7,3.0,3.2,
    3.5,3.8,4.1,4.5,4.8,
    5.1,5.4,5.8,6.2,6.6,
    7.1,7.6,8.0,8.5,9.0
  )
)

ggplot(software, aes(
  x = LinesOfCode,
  y = Bugs,
  size = Developers,
  colour = Complexity
)) +
  geom_point(alpha = 0.85) +
  geom_text(
    aes(label = Team),
    vjust = -0.8,
    size = 2.8,
    show.legend = FALSE
  ) +
  scale_x_log10(
    breaks = c(10000, 100000, 1000000),
    labels = label_number(big.mark = ",")
  ) +
  scale_size_continuous(
```

```
  range = c(2, 9),
  name = "Developers"
 ) +
 scale_colour_viridis_c(
  option = "C",
  name = "Complexity"
 ) +
 labs(
  title = "Software Project Size and Bug Distribution",
  subtitle = "Logarithmic transformation of Lines of Code",
  x = "Lines of Code (log10 scale)",
  y = "Number of Bugs"
 ) +
 theme_minimal()
```



4.2 Explain the aesthetic mappings.

Variable	**Aesthetic**	**Purpose**
Lines of Code	X-axis	Represents project size
Bugs	Y-axis	Represents the number of bugs
Developers	Bubble size	Shows team strength
Complexity	Colour	Indicates project complexity
Team	Text label	Identifies each project team

This integrated scatter plot communicates five project characteristics using position, size, colour, and labels.

4.3 Justify the nonlinear axis transformation.

A base-10 logarithmic transformation is applied to Lines of Code using scale_x_log10().
The transformation is:
\[x'=\log_{10}(x) \]
For example:

- 10,000 lines becomes 4 on the log scale.
- 100,000 lines becomes 5.
- 1,000,000 lines becomes 6.

Equal distances on the transformed axis therefore represent equal multiplicative changes rather than equal absolute differences.
This makes small and large projects easier to display together without allowing the largest projects to dominate the horizontal scale.

4.4 Explain how the transformation improves interpretation over a linear scale.

On a linear scale, a project with 1,800,000 lines of code can dominate the horizontal range, making smaller projects appear compressed toward the origin.
The logarithmic scale:

1. Reduces the visual dominance of very large projects.
2. Makes differences among smaller projects easier to inspect.
3. Preserves the ordering of project sizes.
4. Helps reveal multiplicative patterns between project size and bug counts.
5. Allows bubble size and colour to communicate additional variables.

The logarithmic transformation does not remove the underlying variation; it changes how that variation is displayed.

4.5 Interpret the visualization and justify the design.

Team T20 has the highest Lines of Code (1,800,000), the most bugs (720), the largest developer count (60), and the highest complexity score (9.0).
Team T1 has the smallest project (12,000 lines), 35 bugs, five developers, and a complexity score of 2.1.
The data shows that larger projects are associated with higher bug counts, developer strength, and complexity. However, this association does not establish that project size alone causes more bugs.
Conclusion: A logarithmic x-axis provides a clearer representation of highly skewed project sizes while preserving the ability to compare bug counts, developer strength, and complexity.

Question 5: Integrated Visualization of Publicly Listed Companies

Objective

Develop a visualization that communicates Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector using appropriate aesthetic mappings, coordinate systems, scales, and colour palettes.

5.1 R program for the integrated visualization

A bubble scatter plot is suitable because it can display all five variables in a single chart. Revenue is mapped to the x-axis, Profit Margin to the y-axis, Market Capitalization to bubble size, Employee Strength to colour, and Industry Sector to shape.

```
# Question 5.1: Integrated Company Performance Visualization

library(ggplot2)
library(scales)

companies <- data.frame(
  Company = paste0("C", 1:20),
  Sector = c(
    "IT","Banking","Healthcare","Retail","Energy",
    "IT","Banking","Healthcare","Retail","Energy",
    "IT","Banking","Healthcare","Retail","Energy",
    "IT","Banking","Healthcare","Retail","Energy"
  ),
  Revenue = c(
    420,580,720,850,980,
    1150,1320,1480,1650,1820,
    2050,2280,2550,2840,3150,
    3480,3860,4250,4680,5200
  ),
  ProfitMargin = c(
    8.2,15.6,11.4,9.3,18.5,
    20.4,17.8,14.2,10.5,22.1,
    24.5,19.8,16.7,13.4,26.2,
    28.1,21.5,18.3,15.2,30.4
  ),
  Employees = c(
    1500,3200,2100,4200,2800,
    5100,6800,3900,7200,4500,
    8100,9500,6200,10300,7200,
```

```
    11800,13200,8600,14500,16000
  ),
  MarketCap = c(
    8500,12000,9800,7600,14500,
    18200,21500,16800,13200,24500,
    28500,32500,24800,19200,38200,
    42500,46800,35600,29800,52000
  )
)

companies$Sector <- factor(companies$Sector)

ggplot(companies, aes(
  x = Revenue,
  y = ProfitMargin,
  size = MarketCap,
  colour = Employees,
  shape = Sector
)) +
  geom_point(alpha = 0.85) +
  geom_text(
    aes(label = Company),
    vjust = -0.9,
    size = 2.8,
    show.legend = FALSE
  ) +
  scale_x_continuous(
    labels = label_number(big.mark = ",")
  ) +
  scale_size_area(
    max_size = 12,
    name = "Market Capitalization"
  ) +
  scale_colour_viridis_c(
    option = "C",
    name = "Employees"
  ) +
  scale_shape_manual(
    values = c(16, 17, 15, 18, 8),
    name = "Industry Sector"
```

```
) +
labs(
  title = "Company Revenue, Profitability and Market Value",
  x = "Revenue",
  y = "Profit Margin (%)"
) +
theme_minimal() +
theme(legend.position = "right")
```



5.2 Explain the aesthetic mappings.

Variable	Aesthetic	Justification
Revenue	X-axis	Supports accurate comparison of revenue
Profit Margin	Y-axis	Shows profitability as a percentage
Market Capitalization	Bubble area	Represents differences in market value
Employees	Colour gradient	Represents workforce size as a continuous variable
Industry Sector	Shape	Distinguishes five discrete sectors
Company	Text label	Identifies each company

Using position for Revenue and Profit Margin supports more accurate comparisons than encoding these values through colour or size alone.

5.3 Justify the coordinate system and scale choices.

The Cartesian coordinate system is appropriate because the purpose is to compare two quantitative financial variables.

- Revenue uses a linear scale because the supplied values range from 420 to 5,200, which is manageable for direct comparison.
- Profit Margin uses a linear percentage scale because the values range from 8.2% to 30.4%.
- Market Capitalization is mapped to bubble area rather than radius, reducing the risk of exaggerating differences in market value.
- Employee Strength uses a continuous colour scale because employee counts are quantitative.
- Industry Sector uses distinct point shapes because sectors are categorical.

A logarithmic transformation is not necessary for these supplied ranges to make the chart readable.

5.4 Explain the choice of colour palette and how it minimizes perceptual bias.

The Viridis colour palette is used for Employee Strength. It provides a perceptually ordered gradient, supporting interpretation of lower and higher employee counts while remaining distinguishable for many users with colour-vision deficiencies.

Industry Sector is encoded using shapes rather than additional colours. This separates the categorical and quantitative information and avoids relying entirely on colour.

Bubble area is used for Market Capitalization because readers generally interpret position more accurately than area. Consequently, Revenue and Profit Margin remain on the axes, where their differences are easier to judge.

5.5 Interpret the visualization and justify its effectiveness.

- Company C20 has the highest Revenue (5,200), Profit Margin (30.4%), Employee Strength (16,000), and Market Capitalization (52,000).
- Company C1 has the lowest Revenue (420), a Profit Margin of 8.2%, 1,500 employees, and Market Capitalization of 8,500.
- Company C16 has Revenue of 3,480 and Profit Margin of 28.1%, indicating strong profitability relative to its revenue.
- Company C19 has Revenue of 4,680 and Profit Margin of 15.2%, illustrating that relatively high revenue does not necessarily imply a similarly high profit margin.

The chart allows readers to compare financial performance, workforce size, market value, and sector simultaneously. The visual mappings remain distinct, and the legends explain the meaning of size, colour, and shape.

Conclusion: The integrated bubble scatter plot effectively communicates five company characteristics while minimizing perceptual bias through positional encoding, area-based size mapping, a perceptually ordered colour palette, and distinct sector shapes.